

# SUSHANT LOKHANDE

San Jose, CA | +1 669-340-6655 | [lokhandesushant014@gmail.com](mailto:lokhandesushant014@gmail.com) | [sushantlokhande.me](https://sushantlokhande.me) |  
[linkedin.com/in/sushantlokhande14](https://linkedin.com/in/sushantlokhande14) | [github.com/sushantlokhande14](https://github.com/sushantlokhande14)

## EXPERIENCE

**San Jose State University** | Graduate Research Assistant, Distributed Systems Aug 2025 to May 2026

- Reduced rebuffering **19%** versus a fixed-rate baseline by building a **Docker**, **WebTransport**, and **Media-over-QUIC** testbed and running controlled adaptation experiments across network and protocol configs.
- Cut experiment setup effort **55%** by automating orchestration, configuration, **telemetry** collection, and analysis, making every experiment sweep reproducible from config.
- Validated **14 interoperability scenarios** across networking, dependency, and **failure-recovery** paths to confirm recovery behavior before large experiment runs.
- Boosted malware classification to **80.2%** across 17 families with a soft-voting **PyTorch** ensemble (CNN, ResNet50, ViT), **2.4 pts** over the best single model at  $p < 0.001$ .

**Turito** | Software Engineering Intern, Backend Reliability Jun 2022 to Oct 2022

- Cut controlled-disruption failures **42%** (3.1% to 1.8%) by hardening **retries**, timeouts, failure recovery, and distributed state transitions.
- Slashed p95 **CPU 22%** by profiling high-frequency media paths and eliminating excess buffer allocation, payload copies, repeated routing lookups, and **lock contention**.
- Reduced production diagnosis time **35%** by adding structured **metrics**, runtime-state visibility, and targeted logs.
- Lowered network egress **14%** by removing redundant media-path processing and simplifying routing and payload handling.
- Debugged **ICE/STUN/TURN** connection setup and degraded-network recovery across a **WebRTC** control plane and **RTP/RTCP** media paths spanning about **12 regions**.

**Trivia Softwares** | Software Engineering Intern Nov 2021 to Jan 2022

- Built a **Rust/Tokio** load-testing framework driving about **25,000** concurrent sessions, exposing lock contention, serialized **WebSocket** writes, and slow session cleanup.
- Raised equivalent-load signaling throughput about **31%** by surfacing concurrency bottlenecks that drove targeted backend fixes.

## TECHNICAL SKILLS

**Languages:** Java, Python, C++17, Rust, SQL, TypeScript

**Backend & Distributed:** REST, gRPC, Kafka, Temporal, WebSockets, asyncio, Tokio, FastAPI, Flask

**Data & Cloud:** PostgreSQL, Redis, SQLite, MongoDB, Docker, AWS, Linux, CI/CD

**Quality:** pytest, load testing, Locust, profiling, structured metrics and logging

## PROJECTS

**Relay, LLM Inference Gateway** | *Python, FastAPI, asyncio, SQLite, SSE*

- Cut p50 latency from **759 ms to 44 ms** over a **4,000**-request load test with an OpenAI-compatible gateway doing exact and **semantic caching**, failover, and rate limiting.

**AutoGrade AI, Multi-Agent Evaluation Platform** | *Python, Temporal, Kafka, gRPC, PostgreSQL, Qdrant*

- Orchestrated **4-dimension rubric grading** (correctness, efficiency, resilience, quality) across multiple LLM agents on **Temporal** and **Kafka**, with confidence gates and human-review queues.

**Portcullis, Policy Gateway** | *Rust*

- Built a **Rust** proxy that centralizes authorization and rate limiting and emits auditable accept, reject, and throttle decisions.

## OPEN SOURCE

- corsairdev/corsair**: 6 merged **webhook signature verification** fixes (Telegram, Teams, SharePoint, Monday, GitLab, Jira) that fail closed
- fission/fission**: 3 merged PRs, Kubernetes event and KEDA object capture in support dumps, removed duplicate package enumeration

## EDUCATION

**San Jose State University** | M.S. in Computer Science | GPA 3.8/4.0 | Spartan Scholar Aug 2024 to May 2026

*Coursework:* Distributed Computing, Advanced Algorithms, Database Systems, Computer Networks, Machine Learning, NoSQL

**University of Mumbai** | B.Tech. in Computer Engineering | GPA 3.9/4.0 Jun 2019 to May 2023

*Research:* malware image-classification ensemble selected for Dr. Mark Stamp's ML textbook; sole-author contactless palmprint verification (0.9975 AUC)